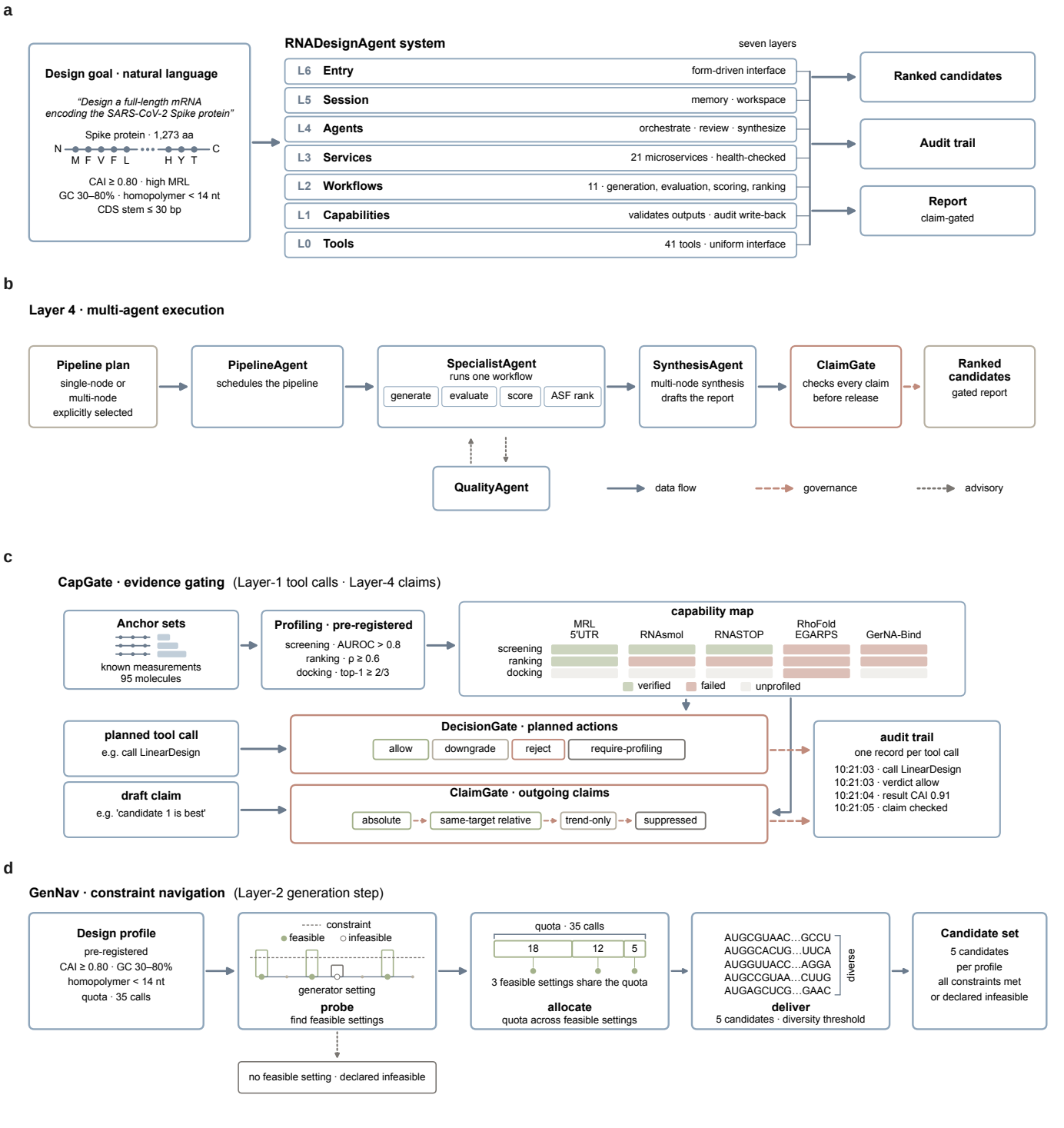

a
Design goal · natural language
"Design a full-length mRNA encoding the SARS-CoV-2 Spike protein"
Spike protein · 1,273 aa
N
C
M F V F L
H Y T
CAI ≥ 0.80 · high MRL
GC 30–80% · homopolymer < 14 nt
CDS stem ≤ 30 bp
RNADesignAgent system
seven layers
L6 Entry
form-driven interface
L5 Session
memory · workspace
L4 Agents
orchestrate · review · synthesize
L3 Services
21 microservices · health-checked
L2 Workflows
11 · generation, evaluation, scoring, ranking
L1 Capabilities
validates outputs · audit write-back
L0 Tools
41 tools · uniform interface
Ranked candidates
Audit trail
Report
claim-gated
b
Layer 4 · multi-agent execution
Pipeline plan
single-node or multi-node explicitly selected
PipelineAgent
schedules the pipeline
SpecialistAgent
runs one workflow
generate
evaluate
score
ASF rank
SynthesisAgent
multi-node synthesis drafts the report
ClaimGate
checks every claim before release
Ranked candidates
gated report
QualityAgent
data flow
governance
advisory
c
CapGate · evidence gating (Layer-1 tool calls · Layer-4 claims)
Anchor sets
known measurements
95 molecules
Profiling · pre-registered
screening · AUROC > 0.8
ranking · ρ ≥ 0.6
docking · top-1 ≥ 2/3
capability map
MRL 5'UTR
RNAsmol
RNASTOP
RhoFold
EGARPS
GerNA-Bind
screening
ranking
docking
verified
failed
unprofiled
planned tool call
e.g. call LinearDesign
DecisionGate · planned actions
allow
downgrade
reject
require-profiling
draft claim
e.g. 'candidate 1 is best'
ClaimGate · outgoing claims
absolute
same-target relative
trend-only
suppressed
audit trail
one record per tool call
10:21:03 · call LinearDesign
10:21:03 · verdict allow
10:21:04 · result CAI 0.91
10:21:05 · claim checked
d
GenNav · constraint navigation (Layer-2 generation step)
Design profile
pre-registered
CAI ≥ 0.80 · GC 30–80%
homopolymer < 14 nt
quota · 35 calls
constraint
feasible
infeasible
generator setting
probe
find feasible settings
no feasible setting · declared infeasible
quota · 35 calls
18
12
5
3 feasible settings share the quota
allocate
quota across feasible settings
AUGCGUAAC…GCCU
AUGGCACUG…UUCA
AUGGUUACC…AGGA
AUGCCGUAA…CUUG
AUGAGCUCG…GAAC
diverse
deliver
5 candidates · diversity threshold
Candidate set
5 candidates
per profile
all constraints met
or declared infeasible